

Vipransh Ojha

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Education

Vellore Institute of Technology

B.Tech in Computer Science and Engineering (AI & ML Specialization)

Bhopal, India

Sep 2023 – May 2027 (Expected)

Technical Skills

- **Languages:** Python, C++, TypeScript, JavaScript, Java, Go (Golang), SQL, HTML5, CSS3
- **Web Dev & Cloud:** React, Node.js, Express.js, Flask, FastAPI, AWS, Docker, Linux, PostgreSQL, MongoDB
- **AI/ML & Data Science:** TensorFlow, PyTorch, Scikit-learn, OpenCV, NLP, Pandas, NumPy, SciPy
- **Tools & Libraries:** Web3 (Ethers.js, Solidity), IPFS, Unity, PyOpenGL, Git, REST APIs

Experience

Software Development Intern

BISAG-N (MeitY, Government of India)

New Delhi, India

May 2025 – Jun 2025

- Deployed a full-stack location intelligence and recommendation system for 500+ users, achieving 85% recommendation accuracy.
- Formulated content-based and collaborative filtering (SVD) algorithms using Python for precise spatial data targeting.
- Architected a highly scalable Flask backend with PostgreSQL, designed to process over 100 queries per second under load.
- Crafted a secure, JWT-authenticated responsive interface with interactive spatial mapping, improving engagement by 25%.

Research Intern (SERB Sponsored Project)

VIT Bhopal University

Bhopal, India

May 2024 – Jun 2024

- Created 'MolSpectra', a molecular simulation and visualization tool for a federally funded research project (Grant No. CRG/2022/002761).
- Engineered a seamless interface integrating 5+ quantum chemistry packages (GAMESS, Open Babel) to automate simulation workflows.
- Constructed the application GUI utilizing PyQt5 and GLEW to streamline the high-fidelity 3D rendering of molecular structures.

Projects

ChronoVault – Decentralized Web3 Storage Protocol

Feb 2026 - Mar 2026

- Architected a zero-knowledge, end-to-end encrypted storage protocol utilizing React, Go (Golang), and IPFS via Pinata.
- Implemented military-grade AES-GCM encryption and sharded chunk distribution, securing file integrity via Merkle Tree verification.
- Integrated Ethereum (Sepolia testnet) Smart Contracts using Solidity and Ethers.js to anchor immutable cryptographic proofs.

Dikastirio – AI Assistant & VR Environment for E-Courtrooms

Nov 2025 – Jan 2026

- Constructed a Unity VR spatial visualization environment, cutting critical document retrieval time by 40%.
- Deployed a local Retrieval-Augmented Generation (RAG) pipeline using LangChain and Llama 3 for secure, offline

document analysis.

- Integrated natural language voice and gesture tracking controls, reducing interaction friction in high-stakes environments by 30%.

SpeechSync – Real-Time Multilingual Translation Pipeline

Apr 2025 - June 2025

- Developed a real-time, low-latency speech-to-speech translation system bridging communication gaps across 4+ languages.
- Engineered a modular Flask backend utilizing OpenAI Whisper (ASR), MarianMT (NMT), and gTTS, achieving 90%+ accuracy.
- Built a responsive browser interface featuring real-time audio waveform visualization and manual/automatic transcription controls.

Automated Dental Forensic Identification System

Aug 2025 – Oct 2025

- Designed an automated computer vision framework for comparative similarity learning on the clinical DENTEX dataset.
- Configured a Siamese Neural Network with an EfficientNet-B0 backbone, optimizing Cosine Similarity metrics and Binary Cross-Entropy.
- Accelerated batch processing throughput, demonstrating a scalable proof-of-concept for matching ante-mortem/post-mortem records.

Certifications

- **Machine Learning Specialization** (Supervised, Unsupervised, & Advanced Algorithms) – *DeepLearning.AI & Stanford University*
- **NPTEL Elite Certifications** (Cloud Computing; Intro to IoT [**Top 2% Topper**]) – *IIT Kharagpur*
- **Generative AI Using IBM Watsonx** – *IBM*
- **Applied Machine Learning in Python** – *University of Michigan*
- **Agentic RAG with LangChain and LangGraph** – *Udemy*
- **AWS Certified Cloud Practitioner Essentials** – *Amazon Web Services (AWS)*

Leadership & Co-Curricular Activities

- **Founding Member & Student Coordinator, Agentic Intelligence Club:** Led university initiatives and fostered a community focused on autonomous AI, workshops, and deep learning applications.
- **Core Organizing Committee, Annual Sports Festival:** Volunteered to manage event logistics, coordinate team schedules, and oversee venue operations for a large-scale university sports fest.
- **Collegiate Athlete:** Represented VIT Bhopal as an active member of the college sports team, demonstrating strong teamwork, discipline, and time management outside of academics.

Achievements

- Placed in the **Top 10** out of 40,000+ registrations in the National Buildathon organized by NASSCOM and Gnani.ai.
- Solved 150+ algorithmic problems on LeetCode.
- Active participant in ongoing coding competitions and hackathons.